

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol

2026-07-17

Source: `results/imo-2026/gpt-5.6-sol/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Approaches tried

- Orbit-displacement approximation — worked. Squared residual identities force arithmetic forward orbits; nearest-integer alignment identifies all positive orbit displacements, and a local cone estimate plus connectedness excludes coexistence of zero and positive displacement.

Current best

The complete classification is

$$\boxed{f(x) = x + c \quad (x > 0), \quad c \geq 0.}$$

Full proof

Set

$$g(t) = f(t) - t.$$

All quantities in the given chain are positive, so squaring preserves both inequalities. Define

$$L(x, y) = (f(x) + y)^2 - 4xf(y), \quad U(x, y) = 2(x^2 + f(y)^2) - (f(x) + y)^2.$$

The hypotheses are exactly

$$L(x, y) \geq 0, \quad U(x, y) \geq 0. \quad (1)$$

Direct expansion gives

$$L(x, y) + U(x, y) = 2(x - f(y))^2, \quad (2)$$

and difference-of-squares factorization gives

$$\begin{aligned} L(x, y) - U(x, y) &= 2((f(x) + y)^2 - (x + f(y))^2) \\ &= 2(g(x) - g(y))(x + y + f(x) + f(y)). \end{aligned} \quad (3)$$

Fix $y > 0$ and substitute $x = f(y)$. By (1) and (2), the two nonnegative numbers $L(f(y), y)$ and $U(f(y), y)$ have sum zero, so both vanish. In particular,

$$(f(f(y)) + y)^2 = 4f(y)^2.$$

Both sides before squaring are positive, hence

$$f(f(y)) + y = 2f(y),$$

which is equivalent to

$$g(f(y)) = g(y). \quad (4)$$

We claim that every forward orbit is arithmetic:

$$f^n(y) = y + ng(y) \quad (n \geq 0). \quad (5)$$

This follows by induction. If it holds for n , repeated use of (4) yields $g(f^n(y)) = g(y)$, and therefore

$$f^{n+1}(y) = f^n(y) + g(f^n(y)) = y + (n + 1)g(y).$$

Since every iterate is positive, (5) implies

$$g(y) \geq 0. \quad (6)$$

Indeed, if $g(y) < 0$, an integer $n > y/(-g(y))$ would give $f^n(y) = y + ng(y) < 0$, a contradiction.

For nonnegative L, U , the triangle inequality gives $|L - U| \leq L + U$. Combining this with (2) and (3) yields the cone estimate

$$|g(x) - g(y)|(x + y + f(x) + f(y)) \leq (x - f(y))^2 \quad (7)$$

for all $x, y > 0$.

We next show that all positive values of g coincide. Suppose

$$a = g(p) > 0, \quad b = g(q) > 0.$$

For every integer $m \geq 0$, let

$$r_m = \frac{q + (m + 1)b - p}{a}.$$

Choose an integer n_m nearest to r_m , so $|n_m - r_m| \leq \frac{1}{2}$. Such an integer exists by taking $n_m = \lfloor r_m + \frac{1}{2} \rfloor$. Since $r_m \rightarrow \infty$, for all sufficiently large m we have $n_m \geq 0$. For those m , put

$$x_m = f^{n_m}(p) = p + n_m a, \quad y_m = f^m(q) = q + mb.$$

By (4),

$$g(x_m) = a, \quad g(y_m) = b, \quad f(y_m) = q + (m+1)b,$$

and the choice of n_m gives

$$|x_m - f(y_m)| = a|n_m - r_m| \leq \frac{a}{2}. \quad (8)$$

Applying (7) to (x_m, y_m) , we obtain

$$|a - b|(x_m + y_m + f(x_m) + f(y_m)) \leq \frac{a^2}{4}. \quad (9)$$

The factor in parentheses is at least $y_m = q + mb$, which tends to infinity. Thus (9) is impossible if $|a - b| > 0$. Hence $a = b$: every positive value assumed by g is the same constant.

It remains to show that zero and a positive displacement cannot coexist. Suppose they do. Then for some $c > 0$,

$$g(t) \in \{0, c\} \quad (t > 0),$$

and both sets

$$Z = \{t > 0 : g(t) = 0\}, \quad P = \{t > 0 : g(t) = c\}$$

are nonempty.

Fix $p \in Z$, so $f(p) = p$. For any $x \in P$, (7) applied to (x, p) gives

$$c(x + p + f(x) + f(p)) \leq (x - p)^2.$$

The left side is greater than cp , so no such x can satisfy $|x - p| < \sqrt{cp}$. Therefore

$$(p - \sqrt{cp}, p + \sqrt{cp}) \cap \mathbb{R}_{>0} \subseteq Z,$$

and Z is relatively open in $\mathbb{R}_{>0}$.

Similarly, fix $q \in P$. For any $y \in Z$, (7) applied to (q, y) gives

$$c(q + y + f(q) + f(y)) \leq (q - y)^2.$$

The left side is greater than cq , so no such y can satisfy $|y - q| < \sqrt{cq}$. Hence

$$(q - \sqrt{cq}, q + \sqrt{cq}) \cap \mathbb{R}_{>0} \subseteq P,$$

and P is relatively open as well.

This partitions the connected interval $\mathbb{R}_{>0}$ into two disjoint, nonempty, relatively open sets, which is impossible by the connectedness theorem for real intervals. Consequently g is constant. By (6), its constant value is some $c \geq 0$, so necessarily

$$f(x) = x + c.$$

Conversely, let $f(x) = x + c$ with $c \geq 0$. This maps positive reals to positive reals, and direct calculation gives

$$(f(x) + y)^2 - 4xf(y) = (x + c + y)^2 - 4x(y + c) = (x - y - c)^2 \geq 0,$$

while

$$2(x^2 + f(y)^2) - (f(x) + y)^2 = 2(x^2 + (y + c)^2) - (x + c + y)^2 = (x - y - c)^2 \geq 0.$$

Because all quantities in the original chain are positive, these squared inequalities are equivalent to the two original inequalities. Thus every $c \geq 0$, including $c = 0$, works, and these are all the solutions. $\square$